The Islamia University of Bahawalpur

Hotel Management System

BY:

Shazma Shafiq ()

BACHELOUR OF COMPUTER SCIENCE

The Islamia University of Bahawalpur

Hotel Management System

BY:

Shazma Shafiq (23447555007)

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DECLARATION

This is to certify that this research work has not been submitted for obtaining similar degree from any other university/college**.**

(**Shazma Shafiq)**

Date :- _____

DEDICATED TO

This project is dedicated to all those who motivated me throughout this journey. I am especially grateful to my teachers for their valuable guidance and to my family for providing continuous support and encouragement. Their contributions played an important role in the successful completion of this work

ACKNOWLEDGEMENT

I would like to express my sincere appreciation for the opportunity to design and develop **Haven AI**, an advanced Hotel Management System that integrates modern web technologies with intelligent automation. This project represents a significant step in applying full-stack development skills to solve real-world challenges in the hospitality industry.

I am grateful to the open-source community and the powerful tools that made this project possible, including React, TypeScript, Tailwind CSS, and shadcn/ui, which enabled the creation of a scalable and user-friendly interface. Special acknowledgment goes to **Lovable** for providing a seamless development and deployment experience that accelerated productivity and collaboration.

This project also benefited from cloud-based services, AI integrations, and modern development practices, allowing **Haven AI** to deliver smart room suggestions, improved guest experience, and efficient hotel operations.

Finally, I acknowledge the continuous learning mindset and dedication that drove this project from concept to implementation. **Haven AI** stands as a reflection of innovation, persistence, and a commitment to building intelligent digital solutions for the future of hospitality.

Revision History

Date	Description	Author	Comments
	Version 1.1	Shazma Shafiq	First Revision

Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Signature	Printed Name	Title	Date
	Mr Jahangir Quershi	Supervisor	

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CHAPTER NO 1
INTRODUCTION

1. INTRODUCTION

1.1 Purpose

The purpose of this Software Design Document (SDD) is to present a clear and detailed description of the Haven AI Hotel Management System. This document explains the objectives, functionality, and operational workflow of the system. It serves as a reference for hotel administrators, staff, developers, and stakeholders by defining how the system is designed to manage hotel operations efficiently.

This document ensures a common understanding of system requirements among all parties involved in the project. It guides the development team in implementing the system correctly and supports future system maintenance and enhancement. By documenting the system in an organized and structured manner, this SDD minimizes confusion and helps maintain consistency in design and functionality.

Additionally, this document explains the motivation behind developing Haven AI, highlighting the challenges of manual hotel management and how this system provides an intelligent digital solution. It also serves as a guide for future developers or hotel management personnel who may update, manage, or expand the system as business needs grow.

1.2 Scope

The scope of the **Haven AI Hotel Management System** includes the automation and management of core hotel operations through a centralized digital platform. The system is designed to replace traditional manual processes with an efficient, secure, and user-friendly solution.

The system covers the following key functionalities:

- Guest registration and profile management
- Room management and availability tracking
- Room booking and reservation handling
- Check-in and check-out management
- Staff and role management
- Attendance and duty scheduling for hotel staff
- Report generation and analytics

1.3 Overview

The **Haven AI Hotel Management System** is a web-based intelligent platform developed to simplify and optimize hotel administration and guest services. The system integrates modern web technologies with artificial intelligence to support smarter decision-making, such as room recommendations and operational insights.

Haven AI improves communication between hotel staff and management, minimizes paperwork, and ensures accurate record-keeping. The system is designed with a user-friendly interface to enhance usability for both technical and non-technical users.

The system consists of two major components:

1.3.1 User-Facing Frontend

The frontend provides access to hotel staff and authorized users to:

- Manage guest profiles and booking information
- View and update room availability
- Handle check-in and check-out operations
- Generate invoices and billing records
- View schedules and assigned duties
- Receive system notifications and alerts

1.3.2 Administrative Backend

The administrative backend offers complete control over the system's data and configuration. Administrators can:

- Manage rooms, room categories, and pricing
- Control staff accounts and access roles
- Monitor reservations and occupancy status
- Generate financial and operational reports
- Configure system settings and security policies

The backend dashboard presents real-time data in an organized format, allowing quick decision-making and smooth hotel operations.

1.4Reference Material

The following resources were used during the design and development of the Haven AI Hotel Management System:

- React Official Documentation – https://react.dev/
- TypeScript Documentation – https://www.typescriptlang.org/
- Tailwind CSS Documentation – https://tailwindcss.com/
- Supabase Documentation – https://supabase.com/docs
- OpenAI API Documentation – https://platform.openai.com/docs
- SendGrid Documentation – https://docs.sendgrid.com/

1.5 Definitions and Acronyms

Term	Description
AI	Artificial Intelligence – Enables smart automation and recommendations
HMS	Hotel Management System
UI	User Interface – Visual components of the system
UX	User Experience – Overall interaction quality
API	Application Programming Interface
DBMS	Database Management System
SQL	Structured Query Language
HTTP	HyperText Transfer Protocol
HTTPS	Secure version of HTTP
IDE	Integrated Development Environment
URL	Uniform Resource Locator

CHAPTER NO 2

SYSTEM OVERVIEW

SYSTEM OVERVIEW

2.1 Assumptions

The development and successful operation of the **Haven AI Hotel Management System** are based on the following assumptions:

- Hotel administrators and staff have basic knowledge of using computers, mobile devices, and the internet.
- The system will be accessed through modern web browsers such as Google Chrome, Mozilla Firefox, or Microsoft Edge.
- A stable and reliable internet connection will be available to ensure smooth system operation.
- The server hosting the application will remain active, secure, and properly maintained.
- All data entered into the system, including guest details, room information, and payment records, will be accurate and entered only by authorized users.
- User login credentials will be kept confidential and will not be shared with unauthorized individuals.
- The hotel will maintain a functional and secure database for storing records such as reservations, room availability, billing, and staff details.
- Necessary devices such as computers, laptops, tablets, or smartphones will be available for users to access the system.
- The system administrator will regularly manage, monitor, and update the backend to ensure uninterrupted services.
- Third-party libraries, APIs, and frameworks used in the system will remain compatible and updated over time.

2.2 Technology Used

The **Haven AI Hotel Management System** is developed using modern and scalable technologies to ensure performance, security, and maintainability.

Backend Technologies

- **Supabase** – Backend-as-a-Service providing authentication, database, and storage
- **PostgreSQL** – Relational database system for secure and efficient data storage
- **OpenAI API** – Used for AI-powered room recommendations and intelligent suggestions
- **SendGrid API** – Used for sending booking confirmations, invoices, and notifications
- **REST APIs** – For communication between frontend and backend services

Frontend Technologies

- **React.js** – JavaScript library for building dynamic and responsive user interfaces
- **TypeScript** – Ensures type safety and better code maintainability
- **Vite** – Fast development and build tool
- **Tailwind CSS** – Utility-first CSS framework for modern UI design
- **shadcn/ui** – Pre-built, reusable UI components

2.3 Design Languages

The **Unified Modeling Language (UML)** is used to visually represent the design and structure of the **Haven AI Hotel Management System**. UML diagrams help in understanding system behavior, data flow, and interactions between system components.

The following UML diagrams are used during the design and development phases:

- **System Architecture Diagram** – Represents overall system structure
- **Entity Relationship Diagram (ERD)** – Shows database entities and relationships
- **Data Flow Diagrams (DFD)** – Illustrate the flow of data within the system
- **Class Diagrams** – Define system classes and their relationships
- **Activity Diagrams** – Describe workflows of hotel operations
- **Sequence Diagrams** – Show interaction between system components over time
- **State Machine Diagrams** – Represent different states of system entities such as room status

These design models ensure clarity, consistency, and ease of understanding for developers and stakeholders, while supporting future enhancements and system scalability.

CHAPTER NO 3
SYSTEM ARCHITECTURE

SYSTEM ARCHITECTURE

3.1 Architectural Design

The **Haven AI Hotel Management System** follows a **three-tier architecture**:

Architecture Layers:

1. Presentation Layer (Frontend)

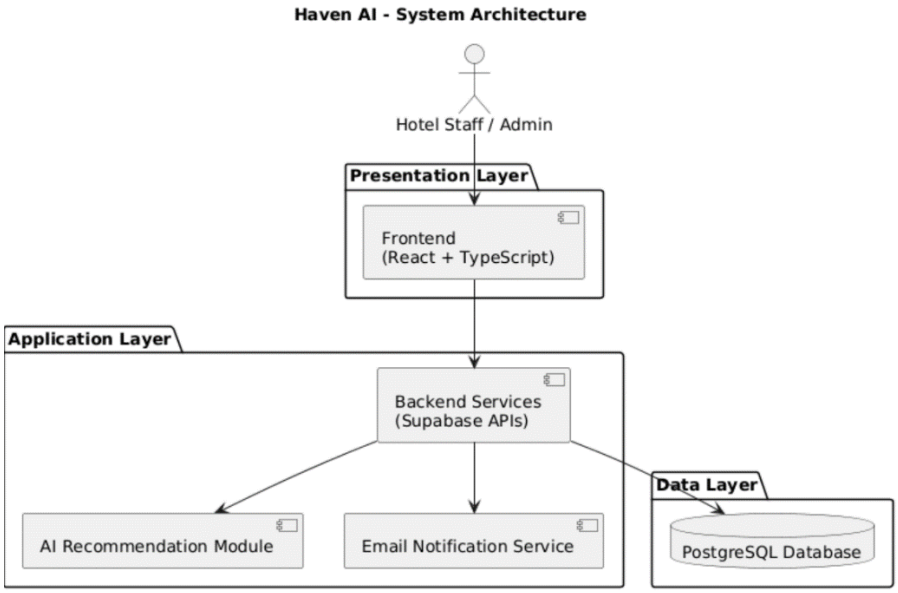
- React + TypeScript Interface
- Dashboards
- Booking Forms
- Admin Panel

2. Application Layer (Backend Services)

- Supabase Backend
- Authentication Module
- Business Logic Module
- AI Recommendation Module
- Email Notification Service

3. Data Layer (Database)

- PostgreSQL Database
- Tables: Guests, Rooms, Reservations, Payments, Staff



3.2 Use Case Diagram

Actors:

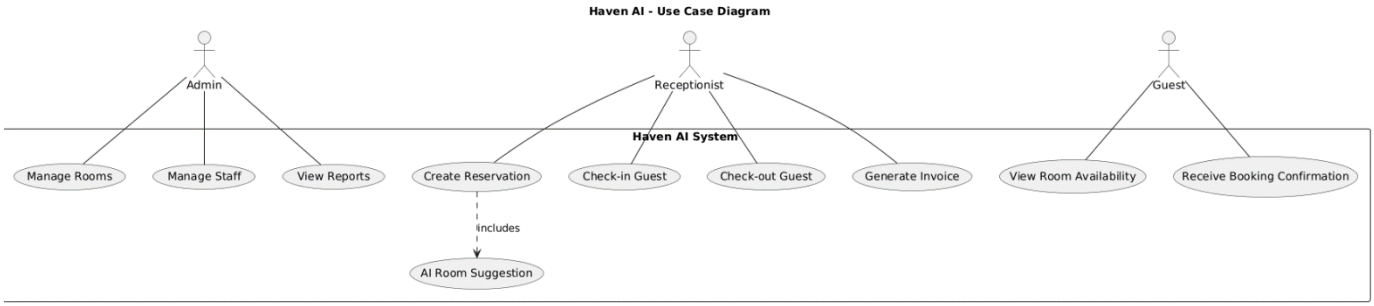
- Admin
- Receptionist
- Hotel Staff
- Guest

Main Use Cases:

- Manage Rooms
- Manage Staff
- Create Reservation
- Check-in Guest
- Check-out Guest
- Generate Invoice
- View Reports
- AI Room Suggestion

Explanation:

The Use Case Diagram represents how different users interact with the system. It shows system functionality from the user's perspective.



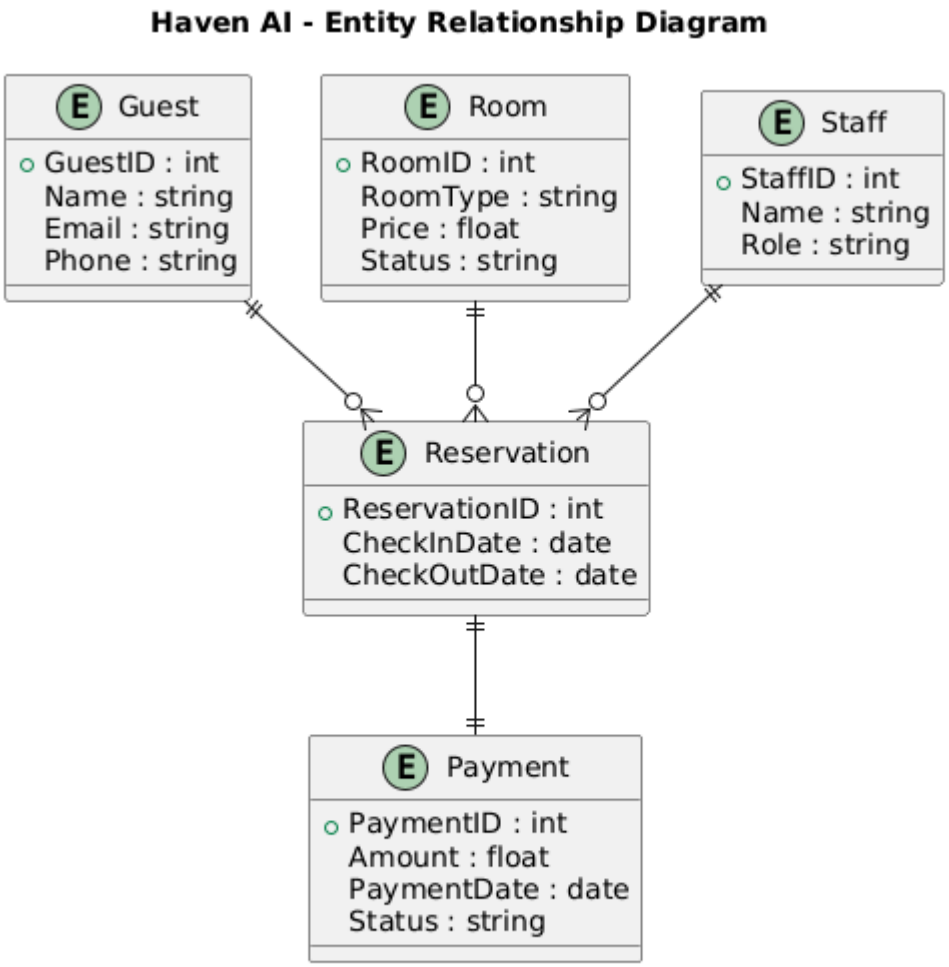
3.3 Entity Relationship Diagram (ERD)

Main Entities:

1. Guest
    - GuestID (PK)
    - Name
    - Email
    - Phone
2. Room
    - RoomID (PK)
    - RoomType
    - Price
    - Status
3. Reservation
    - ReservationID (PK)
    - GuestID (FK)
    - RoomID (FK)
    - CheckInDate
    - CheckOutDate
4. Payment
    - PaymentID (PK)
    - ReservationID (FK)
    - Amount
    - PaymentDate
    - Status
5. Staff
    - StaffID (PK)
    - Name
    - Role

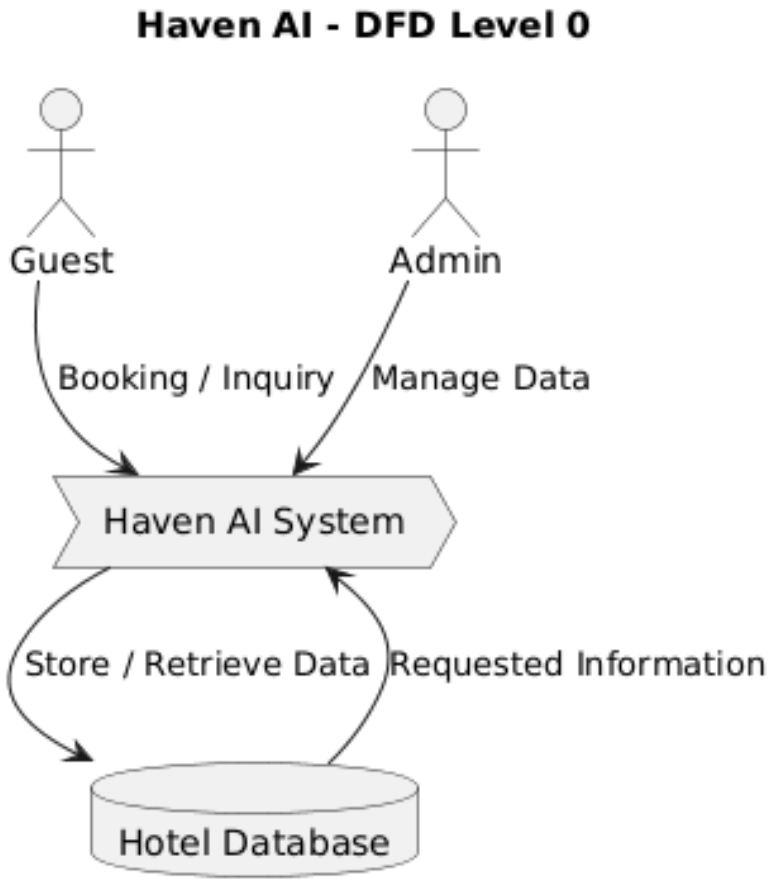
Relationships:

- One Guest can have many Reservations
- One Room can have many Reservations
- One Reservation has one Payment
- Admin manages Staff and Rooms


Haven AI - Entity Relationship Diagram
Guest
GuestID : int
Name : string
Email : string
Phone : string
Room
RoomID : int
RoomType : string
Price : float
Status : string
Staff
StaffID : int
Name : string
Role : string
Reservation
ReservationID : int
CheckInDate : date
CheckOutDate : date
Payment
PaymentID : int
Amount : float
PaymentDate : date
Status : string

3.4 Data Flow Diagram (DFD)

Level 0 (Context Diagram)

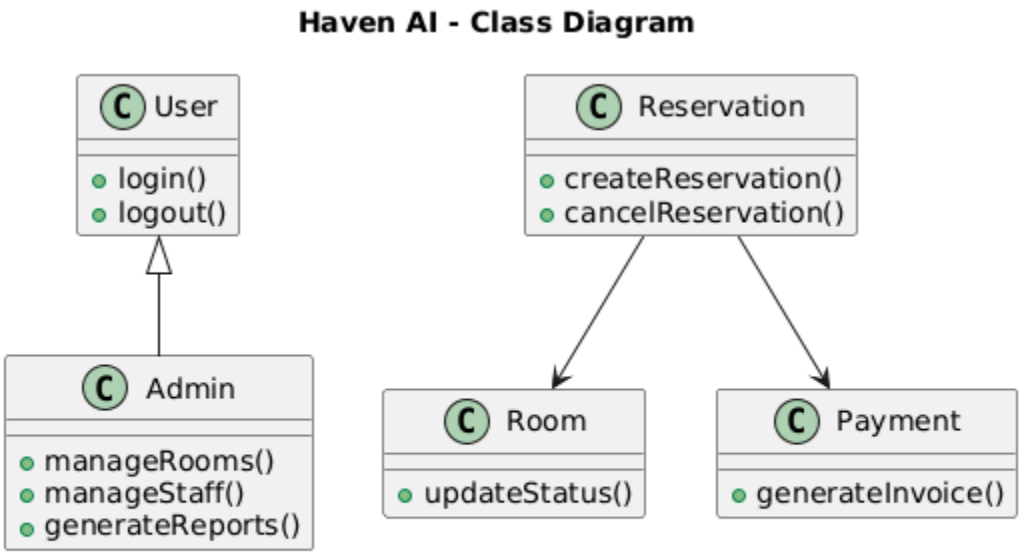


3.5 Class Diagram

Main Classes:

- User
  - login()
  - logout()
- Admin extends User
  - manageRooms()
  - generateReports()
- Reservation
  - createReservation()
  - cancelReservation()
- Room
  - updateStatus()
- Payment
  - generateInvoice()

This diagram shows relationships such as inheritance and associations.

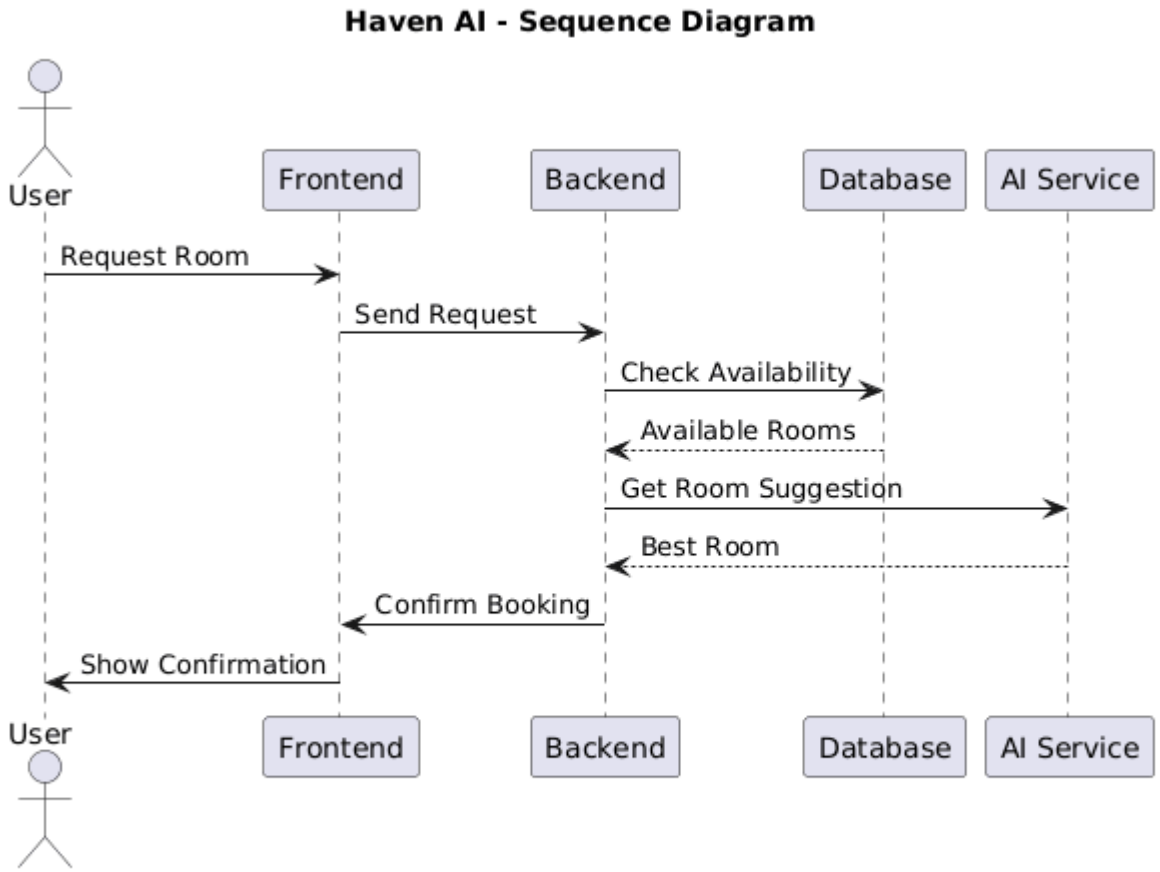


3.7 Sequence Diagram (Reservation Process)

Actors:
User → Frontend → Backend → Database → AI Service

Flow:

1. User requests room
2. Frontend sends request
3. Backend checks availability
4. Database returns data
5. AI suggests best room
6. Backend confirms booking
7. Response sent to user



CHAPTER NO 4
DATA DESIGN

4.1 Introduction

This chapter describes the data design of the **Haven AI Hotel Management System**. Data design defines how data is stored, organized, and managed within the system. A well-structured data design ensures data integrity, security, efficiency, and ease of access. The Haven AI system uses a relational database approach to manage hotel-related data such as guests, rooms, reservations, payments, and staff information.

4.2 Database Design Overview

The Haven AI system uses a centralized relational database to store and manage all hotel operations data. The database is designed to:

- Eliminate data redundancy
- Maintain data consistency
- Ensure data security
- Support efficient querying and reporting

Each entity is represented as a table, and relationships between entities are established using primary keys and foreign keys.

4.3 Entity Description

4.3.1 Guest Table

Attribute	Description
GuestID (PK)	Unique identifier for each guest
Name	Guest full name
Email	Guest email address
Phone	Guest contact number
CNIC / ID	Guest identification number
CreatedAt	Record creation date

4.3.2 Room Table

Attribute	Description
RoomID (PK)	Unique room identifier
RoomNumber	Hotel room number
RoomType	Single, Double, Suite
Price	Room price per night
Status	Available, Reserved, Occupied, Maintenance

4.3.3 Reservation Table

Attribute	Description
ReservationID (PK)	Unique reservation ID
GuestID (FK)	References Guest table
RoomID (FK)	References Room table
CheckInDate	Guest check-in date
CheckOutDate	Guest check-out date
ReservationStatus	Active, Cancelled, Completed

4.3.4 Payment Table

Attribute	Description
PaymentID (PK)	Unique payment ID
ReservationID (FK)	References Reservation table
Amount	Total payable amount
PaymentMethod	Cash, Card, Online
PaymentStatus	Paid, Pending
PaymentDate	Date of payment

4.3.5 Staff Table

Attribute	Description
StaffID (PK)	Unique staff identifier
Name	Staff member name
Role	Admin, Receptionist, Staff
Email	Login email
Phone	Contact number

4.3.6 User Authentication Table

Attribute	Description
UserID (PK)	Unique user ID
StaffID (FK)	References Staff table
Email	User login email
PasswordHash	Encrypted password
Role	Access role
LastLogin	Last login timestamp

4.4 Relationships Between Entities

- One **Guest** can make multiple **Reservations**
- One **Room** can be assigned to multiple **Reservations** (at different times)
- One **Reservation** is associated with one **Payment**
- One **Staff** member can manage multiple **Reservations**
- Each **User** account is linked to one **Staff** record

These relationships are enforced using foreign keys to ensure referential integrity.

4.5 Data Integrity and Constraints

To maintain accuracy and consistency, the following constraints are applied:

- Primary keys ensure unique identification of records
- Foreign keys enforce valid relationships
- NOT NULL constraints prevent missing critical data
- Unique constraints on email and room number
- Check constraints on room status and payment status

4.6 Data Security

The Haven AI system ensures data security by:

- Encrypting sensitive data such as passwords
- Restricting access using role-based permissions
- Securing database communication using HTTPS
- Regular backups of the database

4.7 Data Dictionary (Summary)

Table Name	Purpose
Guest	Stores guest personal information
Room	Stores room details and availability
Reservation	Stores booking information
Payment	Stores billing and payment data
Staff	Stores employee information
UserAuth	Stores login and access credentials

4.8 Conclusion

The data design of the Haven AI Hotel Management System provides a reliable and scalable foundation for managing hotel operations. By using a structured relational database and enforcing data integrity rules, the system ensures accurate data storage, efficient processing, and secure access. This design supports future expansion and integration of advanced features such as AI-driven analytics and reporting.

CHAPTER NO 5
COMMPONENT DESIGN

5.1 Introduction

Component Design describes the internal structure of the **Hotel Management System** and explains how the system is divided into independent, reusable, and manageable software components.

The system is developed using modern full-stack technologies such as:

- **Frontend:** React
- **Backend & Database:** Supabase
- **Email Service Integration:** SendGrid
- **AI Integration:** OpenAI API

Each component is designed to be modular, scalable, secure, and easy to mainta

5.2 Objectives of Component Design

The main objectives of component design are:

- To divide the system into smaller manageable modules
- To improve system maintainability
- To ensure reusability of components
- To enhance system scalability
- To provide secure and efficient communication between modules

5.3 High-Level Component Architecture

The Hotel Management System consists of the following major components:

1. Presentation Layer (Frontend)
2. Application Logic Layer
3. Database Layer
4. External Service Integration Layer
5. Authentication & Security Layer

5.4 Frontend Components (React)

The frontend is developed using **React** and consists of reusable UI components.

5.4.1 Authentication Component

Purpose:
Handles user login, registration, and password recovery.

Subcomponents:

- Login Form
- Registration Form
- Forgot Password Module
- Role-Based Redirection

Functions:

- Validate user credentials
- Communicate with Supabase authentication
- Manage session tokens

5.4.2 Dashboard Component

Purpose:
Provides system overview for Admin and Staff.

Features:

- Total Rooms
- Available Rooms
- Bookings Summary
- Revenue Overview
- Notifications Panel

5.4.3 Room Management Component

Purpose:
Manages hotel rooms.

Subcomponents:

- Add Room
- Update Room
- Delete Room
- View Room Details
- Room Status Control (Available/Booked/Maintenance)

Responsibilities:

- CRUD operations
- Room availability tracking

5.4.4 Booking Management Component

Purpose:
Handles customer reservations.

Subcomponents:

- Create Booking
- Modify Booking
- Cancel Booking
- Booking History

Responsibilities:

- Check room availability
- Calculate total cost
- Generate booking confirmation

5.4.5 Customer Management Component

Purpose:
Stores and manages customer records.

Features:

- Add Customer
- Update Customer
- View Booking History
- Search Customer

5.4.6 AI Room Suggestion Component

Purpose:
Provides intelligent room recommendations.

Integration:
Uses **OpenAI API** for:

- Suggesting best rooms based on budget
- Personalized recommendations
- Improving room descriptions

Workflow:

1. User enters preferences
2. System sends prompt to OpenAI
3. AI returns optimized suggestion
4. Results displayed to user

5.4.7 Email Notification Component

Purpose:
Sends automated emails.

Integration:
Uses **SendGrid API**

Triggers:

- Booking Confirmation
- Cancellation Notice
- Payment Receipt
- Admin Alerts

5.5 Backend Components (Supabase)

The backend is powered by Supabase which provides:

5.5.1 Database Component

Tables:

- Users
- Rooms
- Bookings
- Customers
- Payments

Responsibilities:

- Data storage
- Data integrity
- Relationship management

5.5.2 Authentication Component

Features:

- Role-Based Access Control (Admin/Staff/User)
- JWT-based Authentication
- Secure Password Storage

5.5.3 API Layer

Purpose:

- Connect frontend with database
- Process business logic
- Secure data transactions

5.6 External Integration Components

5.6.1 SendGrid Integration Component

Responsibilities:

- SMTP configuration
- Email template management
- Delivery tracking

5.6.2 OpenAI Integration Component

Responsibilities:

- Prompt construction
- API request handling
- Response parsing
- Error handling

5.7 Component Interaction Diagram (Description)

1. User interacts with React UI
2. React sends request to Supabase
3. Supabase processes database operations
4. If booking confirmed → SendGrid sends email
5. If suggestion requested → OpenAI generates AI recommendation
6. System returns response to frontend

5.8 Security Component Design

Security mechanisms include:

- HTTPS Communication
- JWT Authentication
- Role-Based Access Control
- Input Validation
- SQL Injection Protection
- Environment Variables for API keys

5.9 Error Handling Component

Error handling includes:

- Frontend validation errors
- Backend validation errors
- API timeout handling
- Email failure logging
- AI API fallback mechanism

5.10 Summary

This chapter explained the component-level design of the Hotel Management System.

The system follows a modular architecture using:

- React for UI
- Supabase for backend and database
- SendGrid for email automation
- OpenAI API for intelligent recommendations

CHAPTER NO 6

HUMAN INTERFACE DESIGN

6.1 Introduction

Human Interface Design (HID) focuses on the interaction between users and the **Hotel Management System**. The main objective is to design a user-friendly, intuitive, and responsive interface that enhances user experience while ensuring efficiency, accessibility, and usability.

The system interface is developed using **React**, which enables the creation of interactive, fast, and reusable user interface components. The design follows modern UI/UX principles to minimize user effort and maximize productivity.

6.2 Goals of Human Interface Design

The primary goals of the Human Interface Design are:

- Provide a simple and intuitive user interface
- Reduce user learning time
- Ensure consistency across all screens
- Improve system accessibility and usability
- Support responsive design for different devices
- Minimize user errors through validations and feedback

6.3 User Categories

The Hotel Management System supports the following user types:

1. **Administrator**
2. **Hotel Staff**
3. **Customer/User**

Each user category has a customized interface based on assigned roles and permissions.

6.4 Design Principles

The system interface is designed according to the following principles:

6.4.1 Consistency

- Uniform color schemes
- Standard fonts and icons
- Consistent layout and navigation patterns

6.4.2 Simplicity

- Minimalistic screen design
- Clear labels and instructions

- Avoid unnecessary information

6.4.3 Feedback

- Success and error messages
- Loading indicators
- Confirmation dialogs

6.4.4 Accessibility

- Proper contrast and readable fonts
- Keyboard navigation support
- Screen reader-friendly elements

6.5 Navigation Design

The navigation structure of the system includes:

- Top Navigation Bar
- Sidebar Menu
- Breadcrumb Navigation

Menu Options:

- Dashboard
- Room Management
- Booking Management
- Customer Management
- Payments
- Reports
- Settings
- Logout

6.6 Screen Layout Design

6.6.1 Login Screen

Features:

- Email and Password fields
- Login button
- Forgot Password link
- Role-based authentication

Usability Features:

- Input validation
- Error messages
- Secure authentication via backend

6.6.2 Dashboard Screen

Features:

- Summary cards (Total Rooms, Bookings, Revenue)
- Graphical statistics
- Notifications panel

Design Considerations:

- Clear visual hierarchy
- Real-time updates
- Responsive charts

6.6.3 Room Management Screen

Features:

- Room list with filters
- Add/Edit/Delete room options
- Room availability status

Usability Enhancements:

- Search functionality
- Color-coded room status
- Confirmation dialogs

6.6.4 Booking Management Screen

Features:

- Booking list
- Booking details view
- Cancel or modify booking

User Support:

- Date pickers
- Automatic cost calculation
- Availability indicators

6.6.5 Customer Interface Screen

Features:

- Room search
- Booking form
- Booking history
- Profile management

6.6.6 AI Room Suggestion Screen

Purpose:
Provides intelligent room recommendations using **OpenAI API**.

Features:

- User preference input (budget, dates, room type)
- AI-generated room suggestions
- Improved room descriptions

6.6.7 Email Notification Interface

Purpose:
Displays confirmation messages for emails sent through **SendGrid**.

Notifications Include:

- Booking confirmation
- Cancellation notice
- Payment receipt

6.7 Form Design

Form design focuses on clarity and ease of use:

- Mandatory field indicators
- Inline error messages
- Placeholder text
- Input constraints

6.8 Color Scheme and Typography

6.8.1 Color Scheme

- Primary Color: Blue
- Secondary Color: White
- Accent Color: Green

6.8.2 Typography

- Headings: Bold and large font
- Body Text: Simple and readable font

6.9 Responsiveness and Device Compatibility

The system supports:

- Desktop computers
- Tablets
- Mobile devices

Responsive design is achieved using flexible layouts and media queries.

6.10 Security and Privacy in Interface Design

Security measures integrated into the interface include:

- Hidden password fields
- Session timeout alerts
- Role-based screen access
- Secure logout functionality

6.11 Error Handling and User Guidance

- Clear error messages
- Help tooltips
- Confirmation alerts
- Undo options where applicable

6.12 Summary

This chapter discussed the Human Interface Design of the Hotel Management System. The system provides a user-centered, responsive, and secure interface that ensures ease of use for administrators, staff, and customers.

The interface design, developed using **React**, integrates seamlessly with backend services and external APIs such as **OpenAI** and **SendGrid**, ensuring a modern and intelligent user experience.

CHAPTER NO 7

REQUIREMENTS MATRIX

7.1 Introduction

The Requirements Matrix, also known as the Requirements Traceability Matrix (RTM), is used to ensure that all system requirements of the **Hotel Management System** are properly identified, implemented, and tested.

This matrix establishes a clear relationship between:

- User requirements
- Functional and non-functional requirements
- System components
- Test cases

It helps in tracking project progress and verifying that every requirement is fulfilled.

7.2 Purpose of Requirements Matrix

The main purposes of the Requirements Matrix are:

- To ensure complete coverage of system requirements
- To trace each requirement throughout the development lifecycle
- To verify that requirements are correctly implemented
- To assist in testing and validation
- To reduce the risk of missing or incomplete features

7.3 Requirement Identification

Each requirement is assigned a unique Requirement ID for tracking purposes.

Requirement Types:

- FR → Functional Requirement
- NFR → Non-Functional Requirement

Req ID	Requirement Description	Type	Module / Component	Priority	Status
FR-01	User login and authentication	Functional	Authentication Module	High	Implemented
FR-02	Role-based access control	Functional	Security Component	High	Implemented
FR-03	Manage hotel rooms (Add/Edit/Delete)	Functional	Room Management	High	Implemented
FR-04	Room availability tracking	Functional	Booking Module	High	Implemented
FR-05	Create and manage bookings	Functional	Booking Management	High	Implemented
FR-06	Customer information management	Functional	Customer Module	Medium	Implemented
FR-07	Payment record management	Functional	Payment Module	Medium	Implemented
FR-08	Dashboard analytics and reports	Functional	Dashboard	Medium	Implemented
FR-09	Email notifications for bookings	Functional	Email Notification	Medium	Implemented
FR-10	AI-based room suggestions	Functional	AI Recommendation Module	Medium	Implemented
NFR-01	System performance and fast response	Non-Functional	Entire System	High	Implemented
NFR-02	Data security and encryption	Non-Functional	Security Layer	High	Implemented
NFR-03	System usability and UI consistency	Non-Functional	UI Layer	Medium	Implemented
NFR-04	Scalability for future expansion	Non-Functional	Architecture Design	Medium	Planned
NFR-05	System availability and reliability	Non-Functional	Backend Services	High	Implemented

7.5 Mapping Requirements to Technologies

Requirement Area	Technology Used
User Interface	React
Backend & Database	Supabase
Authentication	Supabase Auth
Email Notifications	SendGrid
AI Recommendations	OpenAI API
Security	JWT, HTTPS

7.6 Requirement Verification Matrix

Req ID	Design Reference	Implementation Module	Test Case ID	Verification Status
FR-01	Component Design	Authentication	TC-01	Passed
FR-03	Component Design	Room Management	TC-03	Passed
FR-05	Component Design	Booking Module	TC-05	Passed
FR-09	Component Design	Email Service	TC-09	Passed
FR-10	Component Design	AI Module	TC-10	Passed
NFR-01	Architecture Design	Backend	TC-11	Passed
NFR-02	Security Design	Security Layer	TC-12	Passed

7.7 Benefits of Requirements Matrix

- Ensures traceability from requirement to testing
- Improves system quality and reliability
- Helps in project monitoring and control
- Supports maintenance and future upgrades

7.8 Summary

This chapter presented the Requirements Matrix for the Hotel Management System. The matrix ensures that all functional and non-functional requirements are properly traced, implemented, and verified.

By maintaining this structured Requirements Matrix, the system achieves better quality assurance, reduced development risk, and higher user satisfaction.